

MACHINE LEARNING [Revised Credit System] (Effective from the academic year 2021-2022) SEMESTER - V			
Subject Code	CSE 3171	IA Marks	50
Number of Lecture Hours/Week	03	Exam Marks	50
Total Number of Lecture Hours	36	Exam Hours	03
CREDITS – 03			
Course objectives: This course will enable students to - To understand the theoretical underpinnings of Machine Learning - To have basic knowledge about the key algorithms and theory that forms the foundation of machine learning. - To have a practical knowledge of machine learning algorithms and methods			
Module -1			Teaching Hours
INTRODUCTION and SUPERVISED LEARNING BASICS: Towards Intelligent Machines, Well-Posed Machine Learning Problems, Examples of Applications in Diverse Fields, Forms of Learning: Supervised Learning, Unsupervised Learning, Reinforcement Learning Learning from Observations, Bias and Variance, Computational Learning Theory, Occam’s Razor Principle and Overfitting Avoidance, Metrics for Assessing Regression Accuracy, Metrics for Assessing Classification Accuracy, An Overview of the Design Cycle and Issues in Machine Learning Text Book 1: Chapter: 1.1 - 1.3, 1.7, 2.1 - 2.4, 2.7 - 2.9			6 Hours
Module -2			
STATISTICAL LEARNING: Bayesian Reasoning: Bayes Theorem, Naïve Bayes Classifier, Bayesian Belief Networks, k-Nearest Neighbour Classifier, Discriminant Functions and Regression Functions, Linear Regression with Least Square Error Criterion, Logistic Regression for Classification Tasks Parametric Methods: Introduction, Maximum Likelihood Estimation, The Bayes Estimator Text Book 1: Chapter: 3.3 - 3.7 Text Book 2: Chapter: 4.1, 4.2, 4.4			8 Hours
Module – 3			
SUPPORT VECTOR MACHINES: Introduction, Linear Discriminant Functions for Binary Classification, Perceptron Algorithm, Linear Maximal Margin Classifier for Linearly Separable Data, Linear Soft Margin Classifier for Overlapping Classes,			6 Hours

Regression by Support Vector Machines: Linear Regression, Nonlinear Regression	
Text Book 1: Chapter: 4.1 - 4.5, 4.8	
Module - 4	
DATA CLUSTERING: Unsupervised Learning, Engineering the Data, Overview of Basic Clustering Methods, K-Means Clustering, Expectation-Maximization Algorithm and Gaussian Mixtures Clustering, Hierarchical Clustering	7 Hours
Text Book 1: Chapter: 7.1 - 7.4, 7.6 Text Book 2: Chapter: 7.8	
Module - 5	
DECISION TREES: Introduction, Example of a Classification Decision Tree, Measure of Impurity for Evaluating Splits in Decision Trees, ID3, C4.5, and CART Decision Trees, Pruning the Tree, Strengths and Weaknesses of Decision Tree Approach	7 hours
Text Book 1: Chapter: 8.1 - 8.6	
Module-6	
COMBINING MULTIPLE LEARNERS: Rationale, Generating Diverse Learners, Voting, Error-Correcting Output Codes, Bagging, Boosting.	2 hours
Text Book 2: Chapter 17.1, 17.2, 17.4 - 17.7	
Course outcomes:	
After studying this course, students will be able: 1. To explain basic concepts of machine learning and supervised learning 2. To apply statistical methods for classification and inference 3. To use support vector machines for classification and regression 4. To apply unsupervised learning methods for clustering 5. To use decision trees for classification	
Text Books: 1. M. Gopal, <i>Applied Machine Learning</i>, McGraw Hill Education, 2018 2. Ethem Alpaydin, <i>Introduction to Machine Learning</i>, 2nd edition, MIT Press. 2010.	
Reference Books: 1. Peter Harrington, <i>Machine Learning in Action</i>, Manning Publications, 2012. 2. Andreas C. Müller & Sarah Guido, <i>Introduction to Machine Learning with Python</i>, O'Reilly Media Inc., 2017 3. Tom M Mitchell, <i>Machine Learning</i>, McGraw Hill, 2017	